Semi-prime factorization – Analysis of Fermat's Factorization method

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Foreword

I am self-taught without formal math education as I dropped out of high-school. This represents 3 years of research and teaching myself mathematics.

Before this I was a self-taught software vulnerability researcher who briefly worked at companies such as Microsoft and Trend Micro and has well over 50+ CVEs in software such as Windows OS, OpenSsl and Adobe Reader attributed to their name (often under the handle SandboxEscaper or Polar Bear).

No AI was used for this research, I have only used it in the last couple of weeks to review my paper and get feedback and occasionally to produce code for well documented mathematical functions.

This paper mainly illustrates findings by example and companion code rather than rigorous mathematical notation and proofs. So perhaps rather than a formal academic paper, view this as a mathematical exploration by an enthusiast with prior experience in IT security.

To do: Add references and closing thoughts and clean everything up.

Chapter I – Fermat's Factorization method

In its most basic form the procedure for Fermat's Factorization method is as follows (where N is the semi-prime being factored).

1. Calculate $\lceil\sqrt{N}\rceil$
2. Starting from $x = \lceil\sqrt{N}\rceil$, calculate $f(x) = x^2 - N$ in a loop, incrementing x.
3. If $f(x)$ evaluates to a square, y^2, compute $GCD(x+y, N)$ and $GCD(x-y, N)$

It is plain to see why this works, because if $x^2 - N = y^2$ then it also follows that $N = x^2 - y^2 = (x-y)(x+y)$ hence anytime we have $x^2 \equiv y^2 \pmod N$ where $x \not\equiv \pm y \pmod N$ the factors can be computed via GCD

Taking $N = 4387$ as an example:

Step 1 (Calculate $\lceil\sqrt{N}\rceil$)

$\lceil\sqrt{4387}\rceil = 67$

Step 2 (Starting from $x = \lceil\sqrt{N}\rceil$, calculate $f(x) = x^2 - N$):

67^2–$4387 = 102$ (not a square)

$68^2 – 4387 = 237$(not a square)
$69^2 – 4387 = 374$ (not a square)
$70^2 – 4387 = 513$ (not a square)
$71^2 – 4387 = 654$ (not a square)
$72^2 – 4387 = 797$ (not a square)
$73^2 – 4387 = 942$ (not a square)
$74^2 – 4387 = 1089 = 33^2$ (square)

Step 3 (If the result is also a square, calculate the GCD on the difference.):

$74^2 \equiv 33^2 \pmod{4387}$

$GCD(74 + 33, 4387) = 107$
$GCD(74 – 33, 4387) = 41$

Both the Quadratic Sieve and Number Field Sieve algorithms (to do: insert references) are expansions on this method.

Chapter II – Analysis

a. Binomial expansion

As seen in chapter 1 the goal is to find the following numbers that we can use to take the GCD with:

$GCD(x + y, N) = q$
$GCD(x – y, N) = p$

x and y can thus be rewritten as the differences of the factors of N divided by 2, $(p + q)/2$ or $(p – q)/2$.
In addition, many such possible x and y solutions exist created by $(ap + bq)/2$ and $(ap – bq)/2$ where a and b are multipliers.
In practice we will not know the factors p and q, but treating them as unknowns we can calculate residues in Z/ℓ (for a prime ℓ) for them using the framework we're establishing here.

If $a = 13$, $b = 5$ and $N = 4387$ (and thus factors as $p = 41$ and $q = 107$)

We get:

$(13 \times 41 + 5 \times 107)/2 = 534$
$(13 \times 41 – 5 \times 107)/2 = –1$

And we can see this creates the following expression when we square 534 $\pmod N$:

$534^2 – 4387 \times 13 \times 5 = 1^2$

One way to represent x and y is through binomial expansion.
I will shorten binomial expansion as $BE(x – y)^e$.
For example $BE(x – 74)^2 = x^2 – 148x + 5476$.

Substituting the binomial term x in $BE(x – 74)^e$, with for example 33, we get:

$BE(33 – 74)^1 = 33 – 74 = –41$
$BE(33 – 74)^2 = 33^2 – 148 \times 33 + 5476 = –41^2$
$BE(33 – 74)^3 = 33^3 – 222 \times 33^2 + 16428 \times 33 – 405224 = –41^3$
$BE(33 – 74)^4 = 33^4 – 296 \times 33^3 + 32856 \times 33^2 – 1620896 \times 33 + 29986576 = –41^4$

Conversely we also see the following occurring if we substitute in the factor as root instead (or swapping a binomial term with the binomial value):

$BE(41 – 74)^1 = 41 – 74 = –33$
$BE(41 – 74)^2 = 41^2 – 148 \times 41 + 5476 = –33^2$
$BE(41 – 74)^3 = 41^3 – 222 \times 41^2 + 16428 \times 41 – 405224 = –33^3$
$BE(41 – 74)^4 = 41^4 – 296 \times 41^3 + 32856 \times 41^2 – 1620896 \times 41 + 29986576 = –33^4$

In addition, if we have even exponents we can substitute the constant by a multiple of N and generate a polynomial $g(x)$ to solve for 0.

$BE(41 – 74)^2 = 41^2 – 148 \times 41 + 5476 \to g(x) = 41^2 – 148 \times 41 + 4387 = 0$
$BE(41 – 74)^4 = 41^4 – 296 \times 41^3 + 32856 \times 41^2 – 1620896 \times 41 + 29986576 \to g(x) = 41^4 – 296 \times 41^3 + 32856 \times 41^2 – 1620896 \times 41 + 4387 \times 6565 = 0$

b. The discriminant

Of-course we cannot blindly perform binomial expansion on random numbers hoping to find a difference of squares. We will need to construct a way to sieve this.

Take for example the following binomial expansion $BE(x – 74)^2 = x^2 – 148x + 5476$ and on this quadratic polynomial , we substitute the constant with N (for example $N = 4387$):

$g(x) = x^2 – 148x + 4387$

If we now calculate the discriminant of $g(x)$, $DISC(g(x)) = 148^2 – 4 \times 4387 = 66^2$, we see that the result is also square.

And we now have the square relation $148^2 \equiv 66^2 \pmod{4387}$, where $GCD(148–66, 4387) = 41$.

An observation:

From this we can make the following generalization that we can derive the factorization of a semi-prime N, if we find a quadratic polynomial with a constant set to N, or a multiple there-of, and whose discriminant yields a square. And from this it is also true that the discriminant would yield a quadratic residue modulo any prime if the discriminant is square in Z. Hence we can use this property to calculate residues in Z/ℓ where valid solutions in Z/ℓ are a quadratic residue and discard residues for quadratics where this is not the case. Now it is also true, that if we find a polynomial whose discriminant is square in Z then it must have a multiple of the factor of N as root for polynomials g(x) at degree 2. That relation between polynomial and discriminant is still something I am researching. Perhaps this will be useful in a type of number field type of construction, hence why I feel it is important to make note of this finding. In addition, it also links everything back to the polynomials generated with binomial expansion.

c. An extended example

Let $N = 4387$
Let binomial $b = x–534$.
Hence $BE(x–534)2 = x^2–1068x + 534^2$

We now need to substitute the constant in the resulting polynomial with a multiple of N.
For example let N multiplier $k = 1$:

$k = 1 \to g(x) = x^2–1068x + 4387$

$DISC(g(x))$ is not a square in Z hence we know $k = 1$ cannot be a valid solution.
However if $k = 65$:

$k = 65 \to g(x) = x^2–1068x + 4387 \times 65$ and $DISC(g(x)) = 2^2$

The discriminant is defined as $DISC(g(x)) = (2t)^2$, where t is the unknown binomial term, we can now substitute this into binomial b.
$b(1) = 1–534 = -533$ and $g(533) = 0$. Since g(x) evaluates to 0 at 533 we know we have a correct solution and we can take the gcd using our binomial terms:

$GCD(1 + 534, 4387) = 107$
$GCD(534–1, 4387) = 41$

In the next chapter we will see how to find these solutions in Z/ℓ

Chapter 3 – Improvements

a. Calculating polynomial residues in $\mathbb{Z}/\ell$

Let us restrict ourselves to binomial expansions to the second degree.

The way we calculate polynomial residues is that we must consider all possible coefficients whose discriminant evaluates to a quadratic residue.

Here we calculate all possible residues for the quadratic polynomial $ax^2 + bx + Nk$ where $N = 4387$ (you can also calculate them for $ax^2 + bx – Nk$ instead, it just changes the side we are calculating in the expression $x^2 = y^2 \pmod{N}$). We will ignore the cases where the leading coefficient is 0, as this would downgrade the degree of our polynomial. And in addition we will also ignore the case where k = 0 for the sake of brevity.

Mod 3

$k = 1, a = 1$ and $b = -1 \rightarrow x = 2$, $k = 1, a = 1$ and $b = -2 \rightarrow x = 1$, $k = 1, a = 2$ and $b = 0 \rightarrow x = 1, 2$

$k = 2, a = 1$ and$b = 0 \rightarrow x = 2, 1$, $k = 2, a = 2$ and $b = -1 \rightarrow x = 1$, $k = 2, a = 2$ and $b = -2 \rightarrow x = 2$

Mod 5

$k = 1, a = 1$ and $b = -2 \rightarrow x = 4, 3$, $k = 1, a = 1$ and $b = -3 \rightarrow x = 2, 1$, $k = 1, a = 2$ and $b = 0 \rightarrow x = 3, 2$,

$k = 1, a = 2$ and $b = -1 \rightarrow x = 4$, $k = 1, a = 2$ and $b = -4 \rightarrow x = 1$, $k = 1, a = 3$ and $b = 0 \rightarrow x = 1, 4$,

$k = 1, a = 3$ and $b = -2 \rightarrow x = 2$, $k = 1, a = 3$ and $b = -3 \rightarrow x = 3$, $k = 1, a = 4$ and $b = -1 \rightarrow x = 3, 1$,

$k = 1, a = 4$ and $b = -4 \rightarrow x = 4, 2$

$k = 2, a = 1$ and $b = 0 \rightarrow x = 4, 1$, $k = 2, a = 1$ and $b = -1 \rightarrow x = 3$, $k = 2, a = 1$ and $b = -4 \rightarrow x = 2$,

$k = 2, a = 2$ and $b = -1 \rightarrow x = 2, 1$, $k = 2, a = 2$ and $b = -4 \rightarrow x = 3, 4$, $k = 2, a = 3$ and $b = -2 \rightarrow x = 3, 1$,

$k = 2, a = 3$ and $b = -3 \rightarrow x = 4, 2$, $k = 2, a = 4$ and $b = 0 \rightarrow x = 2, 3$, $k = 2, a = 4$ and $b = -2 \rightarrow x = 4$,

$k = 2, a = 4$ and $b = -3 \rightarrow x = 1$

$k = 3, a = 1$ and $b = 0 \rightarrow x = 3, 2$, $k = 3, a = 1$ and $b = -2 \rightarrow x = 1$, $k = 3, a = 1$ and $b = -3 \rightarrow x = 4$,

$k = 3, a = 2$ and $b = -2 \rightarrow x = 2, 4$, $k = 3, a = 2$ and $b = -3 \rightarrow x = 1, 3$, $k = 3, a = 3$ and $b = -1 \rightarrow x = 3, 4$,

$k = 3, a = 3$ and $b = -4 \rightarrow x = 2, 1$, $k = 3, a = 4$ and $b = 0 \rightarrow x = 4, 1$, $k = 3, a = 4$ and $b = -1 \rightarrow x = 2$,

$k = 3, a = 4$ and $b = -4 \rightarrow x = 3$

$k = 4, a = 1$ and $b = -1 \rightarrow x = 2, 4$, $k = 4, a = 1$ and $b = -4 \rightarrow x = 1, 3$, $k = 4, a = 2$ and $b = 0 \rightarrow x = 4, 1$,

$k = 4, a = 2$ and $b = -2 \rightarrow x = 3$, $k = 4, a = 2$ and $b = -3 \rightarrow x = 2$, $k = 4, a = 3$ and $b = 0 \rightarrow x = 2, 3$,

$k = 4, a = 3$ and $b = -1 \rightarrow x = 1$, $k = 4, a = 3$ and $b = -4 \rightarrow x = 4$, $k = 4, a = 4$ and $b = -2 \rightarrow x = 1, 2$,

$k = 4, a = 4$ and $b = -3 \rightarrow x = 3, 4$

The problem is, that we can Chinese remainder these together to modulo 15, but then the solution set of possible residues grows quickly (the product of the size of both sets of solutions). So we need to come up with a smart way to generate solutions with this setup.

b. Singular quadratic roots vs non-singular quadratic roots

In the above residue mapping, we will observe that sometimes a set of coefficients will have a single root solution, and sometimes it will have two root solutions.

For example, looking at Mod 5 we see the following two coefficient, root pairings:

$k = 1, a = 4$ and $b = -1 \rightarrow x = 3, 1$ Non-singular roots

$k = 1, a = 3$ and $b = -2 \rightarrow x = 2$ Singular root

Singular roots indicate that the corresponding binomial term has prime ℓ (5 here) as divisor and that the discriminant is also divided by prime ℓ.

We have calculated the above residues using polynomials of the shape $ax^2 + bx + Nk$.

If we calculate the derivative in Z/ℓ we get the residue for linear coefficient b in corresponding to a quadratic with the signs flipped $ax^2 + bx – Nk$ (the other side of the expression $x^2 = y^2 \pmod N$ that we are trying to solve for).

Calculating the derivative for coefficients with a singular root in Z/ℓ gives us a derivative that is 0 in Z/ℓ. Since a singular root means $DISC(g(x)) = 0 \pmod \ell$ and since $DISC(g(x)) = (2t)^2$,we have that $t \equiv 0 \pmod \ell$. (strictly speaking about degree 2 polynomials here).

For example:

Using $k = 1, a = 3$ and $b = -2 \rightarrow x = 2$ we construct the polynomial $g(x) = 3 \times 2^2 – 2 \times 2 + 2$ in $Z/5$.

And we see that the derivative $g(x)' \equiv 2 \times 3 \times 2 – 2 \equiv 0 \pmod 5$ (or in other words 0 in $Z/5$).

This implies that the binomial term y, in binomial b = t – y equals to 0 in Z/5.

Non-singular roots always yield non-zero residues for the binomial terms in Z/ℓ .

TO DO: Slowly adding all chapters... more to follow in the coming days